

The Northeast Asia Semiconductor Triangle

Taiwan's Fabs, Korea's Memory, Japan's Materials

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Executive Summary

Three economies — Taiwan, South Korea, and Japan — form the most consequential industrial cluster in human history: a semiconductor triangle that together produces approximately 80-90% of the world's most advanced logic chips (Taiwan/TSMC), approximately 75% of global DRAM (Korea), approximately 55% of global NAND Flash (Korea/Japan's Kioxia), and a near-total monopoly on the materials and equipment that make all semiconductor manufacturing possible (Japan). The disruption of any corner of this triangle — by natural disaster, geopolitical conflict, trade war, or systemic industrial failure — would cascade through every advanced economy on Earth within months.

This report examines the triangle as an interconnected system: the supply chain dependencies between its three components, the emerging competitive dynamics within the triangle (Japan's Rapidus ambition potentially challenging both Taiwan's logic leadership and Korea's memory dominance), the geopolitical pressures that are simultaneously threatening and reinforcing the triangle's coherence, and the decade-ahead strategic question: will the semiconductor triangle remain concentrated in Northeast Asia, or will US, European, and Chinese policies succeed in globally redistributing semiconductor production?

Part I — The Triangle's Anatomy

Taiwan: The Logic Apex

Taiwan's semiconductor industry is effectively a three-company ecosystem at the leading edge: TSMC (foundry — produces chips designed by fabless companies); TSMC's key advanced packaging partners (ASE, Amkor Taiwan); and the TSMC supply chain that supplies photomasks, chemicals, gases, specialty equipment, and assembly services. But at the triangle's apex is TSMC itself.

TSMC holds approximately 90% of the world's advanced logic foundry market (for nodes 7nm and below) — a share that has no precedent in any other critical global commodity. The second-largest advanced logic foundry, Samsung Foundry, holds approximately 10%. Intel Foundry Services, despite Intel's massive investment commitment, produced no significant external advanced logic volume through 2025.

TSMC's N3E (3nm equivalent) and N2 (2nm, beginning production 2025) nodes represent the frontier of silicon transistor scaling. The N2 node's nanosheet GAAFET (Gate All Around Field Effect Transistor) architecture — the same transistor geometry that Samsung adopted with its SF3 node and that Rapidus is targeting at 2nm — is the inflection point at which transistor scaling begins to hit fundamental physical limits. The density advantages of 2nm over 3nm, and 3nm over 5nm, follow Moore's Law economics that have sustained semiconductor improvement for 60 years.

Korea: The Memory Dominant

South Korea's semiconductor industry holds a duopoly in global DRAM (Samsung at 43%, SK Hynix at 29%) and a leading position in NAND Flash (Samsung at 33%, SK Hynix via Solidigm at approximately 21%). The two Korean DRAM producers together provide approximately 72% of global DRAM supply — a market concentration comparable to TSMC's advanced logic dominance.

Korea's semiconductor geography is concentrated in a remarkably small area: Samsung's primary DRAM and NAND facilities at Pyeongtaek and Hwaseong (south of Seoul, Gyeonggi Province), and SK Hynix's facilities at Icheon (east of Seoul). The entire Korean memory industry — representing 70%+ of global DRAM supply — is concentrated in facilities within approximately 100 km of Seoul. This geographic concentration is Korea's most acute supply chain vulnerability: a single natural disaster or military strike on the Gyeonggi semiconductor cluster would remove 70% of global DRAM supply within days.

Japan: The Materials Chokepoint

Japan's position in the semiconductor triangle is the least visible to casual observers but arguably the most leverage-bearing in a supply chain disruption scenario. Japan controls:

- 70-80% of global advanced photoresist supply (JSR, Shin-Etsu, Tokyo Ohka Kogyo)
- 50% of global 300mm silicon wafer supply (Shin-Etsu Handotai, Sumco)
- 60-70% of global polishing compounds (Fujimi, Cabot Japan)
- 30%+ of global semiconductor equipment (Tokyo Electron, Nikon, Advantest)
- Monopoly on some specialty gases (Kanto Denka, Air Water)

Japan's 2023 export control alignment with the US — restricting exports of specific semiconductor equipment (TEL etching and deposition systems above certain specifications) to China — demonstrated Japan's chokepoint leverage. Chinese semiconductor manufacturers, deprived of specific Japanese process equipment, cannot manufacture certain advanced chips regardless of how much capital they invest in fab construction.

Part II — The Triangle's Internal Dependencies

Taiwan's Dependence on Japanese Materials

TSMC — the triangle's logic apex — is almost entirely dependent on Japanese materials for its leading-edge process. TSMC's 3nm and 2nm process nodes use:

- EUV photoresists from JSR (Japan) — no US or European supplier can substitute
- Silicon wafers from Shin-Etsu Handotai and Sumco (Japan) — these two companies supply the overwhelming majority of TSMC's 300mm wafer inputs
- Multiple process chemicals from Kanto Denka, Stella Chemifa, and other Japanese specialty chemical companies

If Japan's export control regime were extended — hypothetically, in a geopolitical crisis scenario — to restrict photoresist or wafer supply to Taiwan (an implausible scenario given Japan-Taiwan strategic alignment, but illustrative of the dependency) TSMC's leading-edge production would halt within weeks. Japan and Taiwan's semiconductor interdependency is near-total and mutual: Japan's materials industry needs TSMC's technical roadmap to develop next-generation materials (JSR's photoresist R&D is conducted in partnership with TSMC's process engineers); TSMC needs Japan's materials to execute its roadmap.

Korea's Dependence on Japanese Materials

Samsung and SK Hynix's memory fabs share similar dependencies on Japanese materials — with the added factor that Japan imposed actual export controls on three semiconductor materials to Korea in July

2019 (fluorinated polyimide, photoresists, and hydrogen fluoride) during a bilateral trade dispute. The 2019 controls, which Japan framed as export administration measures rather than retaliatory trade sanctions, caused significant concern in the Korean semiconductor industry about the vulnerability of Japanese material dependencies.

Korea responded to the 2019 controls with a systematic effort to diversify material supply away from Japan — supporting domestic Korean material companies, qualifying Belgian, German, and US alternative suppliers, and building emergency stockpiles. By 2025, Korea had materially reduced its dependency on the specific controlled materials — hydrogen fluoride domestic and alternate-source supply increased from approximately 20% to 50%+ of demand. However, the effort also revealed the extreme difficulty of rapidly substituting Japanese semiconductor materials: development and qualification of a new photoresist supplier at leading-edge process nodes takes 2-4 years of technical collaboration.

Part III — The Geopolitical Stress Test

The Taiwan Risk: The Triangle's Achilles Heel

The Taiwan Strait crisis scenario — whatever form it takes, from a naval blockade to an amphibious assault to a missile-only coercive campaign — is the existential threat to the semiconductor triangle. TSMC's Hsinchu, Tainan, and Taichung facilities represent the world's most irreplaceable manufacturing assets. No other country, company, or combination could replicate TSMC's advanced logic manufacturing capability on any timeline shorter than 10-15 years.

The economic consequence of a Taiwan Strait conflict severe enough to disrupt TSMC production has been modelled by multiple institutions. A six-month disruption of TSMC's most advanced nodes would reduce global semiconductor output by an estimated 35-40% — with cascading effects on electronics, automotive, industrial, and defense industries globally. Economic damage in the range of \$1-2 trillion in the first year is consistent with multiple independent estimates.

The US-China Competition: Centrifugal Force on the Triangle

The US-China semiconductor technology war is simultaneously the most significant external force acting on the Northeast Asia semiconductor triangle and the clearest illustration of the triangle's strategic importance. US export controls on advanced semiconductor equipment (BIS Entity List restrictions, FDP rules) are explicitly designed to prevent China from developing leading-edge semiconductor manufacturing capability — which would require purchasing ASML EUV systems (Dutch, but US-IP-content regulated), TSMC-quality photoresists (Japanese), and US-origin semiconductor design software (EDA tools from Cadence and Synopsis).

The triangle's three members have responded to US-China competition in aligned but distinct ways:

- Taiwan: Tightened export controls on chip production equipment, reduced TSMC's China exposure (limiting advanced node production for Chinese customers), and explicitly aligned with US technology access frameworks
- Korea: Extended its US export control compliance (restricting chip equipment to China beyond US requirements), but has been careful to maintain Samsung/SK Hynix's Chinese NAND/DRAM facilities (within the waiver framework the US has granted)
- Japan: Implemented the most aggressive export control expansion after the US — restricting 23 categories of semiconductor equipment beyond pre-existing controls — and committed to the US-Japan-Netherlands trilateral semiconductor equipment export control framework

Part IV — New Competition and the Triangle's Future

China's Self-Sufficiency Programme: SMIC and YMTC

China's response to the semiconductor triangle's dominance — and the US-led export control regime that reinforces it — is the most ambitious industrial policy programme since the Manhattan Project. SMIC (Semiconductor Manufacturing International Corporation) has achieved production of 7nm-equivalent chips using workarounds for the lack of EUV lithography (using multiple ArF immersion passes — technically feasible but economically expensive), demonstrating that China can manufacture advanced chips without EUV given sufficient process ingenuity and willingness to accept higher cost per chip.

YMTC (Yangtze Memory Technologies) achieved 232-layer NAND Flash technology in 2022-2023 — a layer count matching Japan's Kioxia and Samsung's leading NAND products — before US export controls in October 2022 restricted YMTC's access to advanced equipment and essentially froze YMTC's technology development trajectory.

China's self-sufficiency target (70% domestic semiconductor supply by 2025 — a target widely understood to have failed at the advanced level) has been recalibrated to approximately 30-35% overall self-sufficiency by 2025, with advanced nodes remaining import-dependent due to the equipment restriction. The semiconductor triangle's dominance is not existentially threatened by China's domestic programme in any timeframe shorter than 10-15 years — but China's progress in mature nodes (28nm and above) is real and is gradually displacing imports in market segments that do not require cutting-edge process technology.

Conclusion: The Triangle Endures

The Northeast Asia Semiconductor Triangle is the world's most concentrated, most critical, and most geopolitically exposed industrial cluster. It cannot be rapidly duplicated — the combination of TSMC's process IP, Korea's memory manufacturing scale, and Japan's materials ecosystem represents 30-60 years of accumulated industrial learning that cannot be compressed into 5-10-year national semiconductor programmes, however well-funded.

The triangle will face pressure from all directions: US and European CHIPS Act investments intending to diversify advanced semiconductor geography; Chinese investment in domestic alternatives; natural disaster risk (Taiwan earthquakes, Korea floods); and geopolitical crisis scenarios that are impossible to fully discount. But the fundamental economics — the cost advantages of concentrated, scale-efficient manufacturing; the materials ecosystem co-location advantages; the talent concentration — will sustain the triangle's production dominance for the foreseeable future.

What the geopolitical situation has changed is the triangle's self-perception: Taiwan, Korea, and Japan understand their semiconductor assets as strategic national resources requiring government protection, supply chain diversification, and alliance architecture — not merely commercial industry to be left to market forces. That recognition is, in itself, a form of industrial resilience that the triangle's democratic competitors are attempting to replicate.

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All data from SEMI, TSMC, Samsung, Kioxia disclosures, TrendForce, SIA, and cited news sources.